

## TULLINGE, Sweden

WMO: 024690

Lat: 59.1785N

Lon: 17.9093E

Elev: 45

StdP: 100.79

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | -19.0      | -15.3 | -21.5                         | 0.6 | -18.9 | -17.4 | 0.8 | -15.1 | 10.0                  | 0.8  | 9.0 | 1.5  | 0.5                      | 230  | 0.607 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.8                   | 26.4            | 18.4 | 24.6 | 17.6 | 22.8 | 16.5 | 19.7                | 23.9 | 18.6 | 22.7 | 17.6 | 21.7 | 3.5                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.3                            | 13.3 | 21.3 | 17.0 | 12.2 | 20.3 | 16.0 | 11.4 | 19.4 | 56.4          | 24.1 | 52.8 | 22.5 | 49.8 | 21.9 | 23.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 8.2           | 7.2  | 6.4 | DB                         | -23.0 | 28.5               | 4.6 | 1.8                                                | -26.3 | 29.8       | -29.0 | 30.9       | -31.6 | 31.9       | -35.0 | 33.2 | (4) |
| (5)               |      |     | WB                         | -24.4 | 21.0               | 3.6 | 1.6                                                | -26.9 | 22.1       | -29.0 | 23.0       | -31.1 | 23.8       | -33.7 | 25.0 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 6.1   | -3.1  | -2.9  | -0.4  | 4.4   | 9.6   | 13.6  | 16.8  | 15.5  | 11.5  | 6.0   | 2.4   | -0.7 | (6)  |
| (7)  |                                                                   | DBStd       | 7.94  | 5.52  | 4.98  | 4.87  | 3.33  | 3.53  | 2.94  | 2.66  | 2.65  | 3.10  | 3.66  | 3.74  | 5.00 | (7)  |
| (8)  |                                                                   | HDD10.0     | 2030  | 405   | 361   | 324   | 171   | 51    | 5     | 0     | 1     | 21    | 131   | 229   | 333  | (8)  |
| (9)  |                                                                   | HDD18.3     | 4491  | 663   | 594   | 582   | 418   | 271   | 145   | 64    | 94    | 206   | 382   | 479   | 591  | (9)  |
| (10) |                                                                   | CDD10.0     | 605   | 0     | 0     | 0     | 2     | 38    | 112   | 209   | 172   | 65    | 7     | 0     | 0    | (10) |
| (11) |                                                                   | CDD18.3     | 24    | 0     | 0     | 0     | 0     | 0     | 2     | 16    | 7     | 0     | 0     | 0     | 0    | (11) |
| (12) |                                                                   | CDH23.3     | 298   | 0     | 0     | 0     | 0     | 7     | 36    | 200   | 55    | 1     | 0     | 0     | 0    | (12) |
| (13) |                                                                   | CDH26.7     | 43    | 0     | 0     | 0     | 0     | 0     | 1     | 36    | 6     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 3.1   | 3.1   | 3.0   | 3.2   | 3.1   | 3.2   | 3.2   | 3.0   | 2.7   | 3.0   | 3.1   | 3.2   | 3.3  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 609   | 44    | 35    | 31    | 35    | 39    | 59    | 67    | 71    | 59    | 57    | 58    | 54   | (15) |
| (16) |                                                                   | PrecMax     | 778   | 85    | 75    | 77    | 90    | 99    | 111   | 126   | 177   | 176   | 119   | 134   | 128  | (16) |
| (17) |                                                                   | PrecMin     | 429   | 7     | 8     | 7     | 6     | 13    | 19    | 17    | 16    | 8     | 8     | 10    | 11   | (17) |
| (18) |                                                                   | PrecStd     | 69    | 21    | 13    | 16    | 23    | 19    | 27    | 32    | 40    | 38    | 26    | 28    | 26   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 7.5   | 8.0   | 13.6  | 18.5  | 24.2  | 26.5  | 29.7  | 27.2  | 22.1  | 15.7  | 11.2  | 10.4 | (19) |
| (20) |                                                                   |             | MCWB  | 5.8   | 5.2   | 7.9   | 10.5  | 15.3  | 17.5  | 18.9  | 19.7  | 16.7  | 12.7  | 9.5   | 8.7  | (20) |
| (21) |                                                                   | 2%          | DB    | 5.7   | 6.0   | 10.4  | 16.2  | 21.8  | 24.4  | 27.6  | 24.8  | 20.2  | 14.1  | 9.7   | 7.8  | (21) |
| (22) |                                                                   |             | MCWB  | 4.4   | 4.0   | 5.8   | 9.4   | 14.0  | 16.5  | 18.8  | 18.8  | 15.4  | 12.1  | 8.6   | 6.4  | (22) |
| (23) |                                                                   | 5%          | DB    | 4.6   | 4.6   | 8.3   | 13.9  | 19.8  | 22.5  | 25.6  | 22.9  | 18.6  | 12.8  | 8.6   | 6.4  | (23) |
| (24) |                                                                   |             | MCWB  | 3.5   | 3.2   | 4.9   | 8.5   | 13.2  | 15.2  | 18.2  | 17.6  | 14.7  | 11.1  | 7.7   | 5.3  | (24) |
| (25) |                                                                   | 10%         | DB    | 3.5   | 3.4   | 6.1   | 11.6  | 17.3  | 20.7  | 23.7  | 21.4  | 17.2  | 11.7  | 7.7   | 5.3  | (25) |
| (26) |                                                                   |             | MCWB  | 2.6   | 2.4   | 3.5   | 6.8   | 11.6  | 14.4  | 17.3  | 16.7  | 13.8  | 10.3  | 6.8   | 4.4  | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 6.1   | 6.3   | 8.1   | 11.6  | 16.6  | 19.0  | 21.5  | 21.8  | 17.3  | 13.6  | 9.6   | 8.9  | (27) |
| (28) |                                                                   |             | MCDWB | 6.9   | 7.5   | 12.2  | 17.0  | 22.3  | 24.1  | 25.6  | 25.6  | 20.3  | 14.8  | 10.3  | 10.1 | (28) |
| (29) |                                                                   | 2%          | WB    | 4.7   | 4.4   | 6.3   | 10.0  | 15.0  | 17.3  | 20.0  | 19.6  | 16.1  | 12.3  | 8.7   | 6.8  | (29) |
| (30) |                                                                   |             | MCDWB | 5.5   | 5.4   | 9.6   | 15.1  | 19.9  | 22.6  | 24.8  | 22.8  | 19.1  | 13.6  | 9.4   | 7.5  | (30) |
| (31) |                                                                   | 5%          | WB    | 3.7   | 3.4   | 5.0   | 8.7   | 13.6  | 16.1  | 18.9  | 18.4  | 15.3  | 11.4  | 7.7   | 5.5  | (31) |
| (32) |                                                                   |             | MCDWB | 4.4   | 4.4   | 7.8   | 13.1  | 18.5  | 21.0  | 23.5  | 21.6  | 17.8  | 12.6  | 8.4   | 6.2  | (32) |
| (33) |                                                                   | 10%         | WB    | 2.6   | 2.4   | 3.9   | 7.3   | 12.3  | 15.0  | 18.0  | 17.5  | 14.3  | 10.4  | 6.8   | 4.5  | (33) |
| (34) |                                                                   |             | MCDWB | 3.4   | 3.4   | 5.8   | 11.0  | 16.6  | 19.4  | 22.9  | 20.5  | 16.5  | 11.5  | 7.5   | 5.3  | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 5.9   | 6.8   | 8.8   | 11.2  | 11.4  | 10.8  | 10.8  | 10.4  | 9.5   | 7.3   | 5.0   | 5.2  | (35) |
| (36) |                                                                   |             | MCDBR | 5.2   | 6.8   | 11.5  | 15.7  | 15.3  | 14.5  | 14.4  | 12.7  | 11.7  | 8.0   | 6.0   | 5.1  | (36) |
| (37) |                                                                   |             | MCWBR | 5.0   | 5.5   | 7.7   | 9.6   | 8.7   | 7.5   | 6.6   | 7.2   | 7.5   | 6.2   | 5.6   | 4.8  | (37) |
| (38) |                                                                   | 5% WB       | MCDWR | 5.0   | 5.9   | 10.6  | 14.2  | 13.2  | 12.6  | 11.7  | 10.9  | 10.6  | 7.0   | 5.7   | 5.0  | (38) |
| (39) |                                                                   |             | MCWBR | 4.9   | 5.0   | 7.4   | 9.0   | 8.1   | 7.0   | 6.1   | 6.6   | 7.1   | 5.8   | 5.4   | 4.8  | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.284 | 0.321 | 0.339 | 0.376 | 0.372 | 0.362 | 0.390 | 0.381 | 0.348 | 0.337 | 0.310 | 0.269 | (40) |      |
| (41) |                                                                   | taud        | 2.227 | 2.292 | 2.358 | 2.325 | 2.401 | 2.461 | 2.404 | 2.423 | 2.498 | 2.419 | 2.260 | 2.148 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 522   | 690   | 799   | 826   | 858   | 873   | 835   | 815   | 789   | 671   | 478   | 391   | (42) |      |
| (43) |                                                                   | Edh at Noon | 44    | 69    | 87    | 106   | 105   | 101   | 105   | 96    | 76    | 61    | 43    | 33    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 0.30  | 0.89  | 2.40  | 4.01  | 5.32  | 5.90  | 5.44  | 4.30  | 2.80  | 1.23  | 0.40  | 0.20  | (44) |      |
| (45) |                                                                   | RadStd      | 0.04  | 0.15  | 0.27  | 0.38  | 0.45  | 0.33  | 0.51  | 0.38  | 0.31  | 0.20  | 0.09  | 0.03  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |      | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |      | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j)  | (k)         | (l)     | (m)     | (n)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A  | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (52 neighbors) |  | +0.36 | N/A     | N/A    | N/A   | N/A     | N/A   | -116 | -145        | N/A     | N/A     | N/A     |

Nomenclature: See separate page